

A Lightweight Landmark-Based Model for Bahasa Lip-Reading Using Attention-BiLSTM

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Abstract

Background: Most lip-reading studies primarily utilize image-based representations of lip movements, offering extensive visual data while imposing significant computational burdens. Lip landmark-based representations are still not well understood, even though they could be a better way to describe lip dynamics in a smaller and more efficient way. This limitation is even more apparent in Bahasa lip-reading research, where there are few studies and computationally efficient solutions remain essential.

Objective: This study examines the shortcomings of image-based lip-reading methods by leveraging lip landmarks as a concise and computationally efficient input representation. The proposed method is tested on the IndoLR open dataset, which contains video data of lip-reading in Bahasa.

Methods: In this study, video sequences were transformed into coordinate-based landmark data to minimize computational demands while preserving critical information regarding lip dynamics. An attention-based BiLSTM model was trained to group 10 word classes and 4 phrase classes using this dataset.

Results: The model achieved accuracies of 93.03% for word classification and 95.18% for phrase classification. The approach also maintained high efficiency, with average inference times of 0.000530 and 0.011366 s per sample and computational costs of only 0.01 and 0.14 GFLOPs, respectively.

Conclusion: These results show how well lip landmarks can be combined with a lightweight deep learning model with very few resources. This study makes a significant contribution to research on lip-reading in Bahasa and lays the groundwork for future studies that will use larger and more diverse datasets.

Keywords: Attention, BiLSTM, Bahasa, Landmark, Lip-Reading

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I. INTRODUCTION

Lip-reading is the ability to understand what someone is saying by watching their lips' movements. This skill is especially helpful for people who are deaf or have partial hearing loss. Hearing loss makes it much harder to communicate clearly. Hearing aids are getting better quickly, but many people still have trouble understanding speech, especially when there is a lot of noise [1], [2]. Lip-reading technology is a useful communication tool for people who are unable to hear well. This technology can recognize and understand lip movements without the need for spoken words. Then, it can turn them into text or synthetic speech. Therefore, it could significantly improve the communication skills of people who cannot speak and are unfamiliar with sign language. Lip-reading systems can also make people less dependent on personal translators, which encourages communication independence.

The progress of lip-reading technology is presently more evident in widely spoken languages, such as English, primarily due to comprehensive and high-quality data accessibility. This data supports the training of automatic lip movement reading models and contributes to the field of natural language processing (NLP). The Lip Reading in the Wild (LRW) and Lip Reading Sentences 2 (LRS2) datasets have been very important in making good models for English [3], [4], [5], [6], [7]. Unlike the substantial corpus of research on English and other extensively studied languages, investigations into lip movements in Bahasa are scarce, predominantly concentrating on vowels and utilizing relatively small datasets [1], [8], [9], [10]. Research on the visual classification of phonemes or visemes in this language is still sparse, resulting in a substantial gap in the literature. Bahasa is a phonemic orthographic language, which means that the spelling and pronunciation are very similar. This is different from English but similar to languages like Deutsch and Español [11], [12]. This characteristic makes it particularly suitable for visual speech

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recognition; however, research exploiting this advantage is still limited. The gap is interesting because Bahasa is spoken by approximately 260 million people and is also a regional lingua franca for people in nearby countries. Furthermore, the language has achieved international recognition as one of the official languages sanctioned by UNESCO at the United Nations. Therefore, systematic studies that analyze lip gesture patterns in Bahasa are needed not only because Bahasa is widely spoken but also because studying visual speech patterns in a phonemic language provides broader insights into the generalizability of lip reading models across languages [13]. The recent availability of an open audiovisual dataset in Bahasa presents a timely opportunity to enhance research on lip gesture recognition and its prospective applications in nonverbal communication [1], [14]. However, compared with the vast datasets and resources accessible for English, materials facilitating lip-reading research in Bahasa are significantly constrained.

Most of the available datasets are video recordings of people saying certain words or phrases. Then, the video content is changed into a format that can be used for training. The literature review shows that preparing and training lip movement recognition data can be slow, especially when hundreds or thousands of image samples are used, which can be difficult on computers. Conversely, a dataset containing insufficient images may result in suboptimal accuracy [14], [15]. Lip-reading models typically combine diverse artificial neural network architectures to discern image patterns, particularly convolutional neural networks (CNN), in conjunction with architectures tailored for sequential data processing, such as long short-term memory (LSTM) networks. Some studies have also added an attention layer to the data to improve the output by focusing on important features [2], [16], [17], [18].

Previous studies utilized the Mediapipe framework to delineate the lip region and subsequently examined lip movements through a dataset of lip clippings via CNN, LSTM, and Bidirectional LSTM (BiLSTM) networks [14], [18]. However, these landmarks are usually employed as supplemental features instead of the main dataset. This is because landmark extraction requires an additional processing step that relies on reliable face detection and alignment. As a result, many previous studies have preferred end-to-end image-based pipelines, and the use of landmark trajectories as major inputs remains unexplored [19]. The method used in this study differs from end-to-end lip-reading models. Instead of jointly optimizing all components, this study separates lip landmark extraction and temporal modeling into distinct stages. While end-to-end approaches simplify training by simultaneously optimizing the entire pipeline, the learned representations are often implicit and difficult to analyze. In contrast, the proposed multistage approach provides clearer control over each processing stage, enabling explicit modeling of lip movements through landmark trajectories. This design facilitates the focused analysis of lip motion in landmark-based lip-reading research.

This study proposes a lightweight landmark-based model to improve visual speech recognition efficiency by utilizing lip-landmark coordinates rather than full-frame images. Previous research indicates that lip-reading based on localized features is computationally lighter than processing entire image sequences [16], [17]. Although many recent studies have used full-face or full-frame images to enable end-to-end learning and eliminate an additional landmark extraction stage, this approach frequently results in higher processing costs due to the redundant visual information present in raw images. Based on this premise, the present work employs a coordinate-based representation to significantly reduce computational load and processing time while retaining essential information about lip dynamics. To support this approach, an effective Deep Learning (DL) model is developed to accurately classify lip-movement sequences. In contrast to earlier studies that relied on raw image inputs or complex architectures, such as the long-term recurrent convolutional network (LRCN), the proposed method extracts landmark coordinates specifically from the face and lip regions [14], [20], [21]. A quantitative comparison with the LRCN-3CONV model in terms of parameter count and computational cost under identical experimental settings is presented in the Results section. This targeted feature representation minimizes data complexity while preserving the critical motion cues required for robust VSR.

The adoption of an attention-based BiLSTM model, which captures temporal dependencies and emphasizes informative lip-movement features, is a key contribution of this study. The experimental results demonstrate improvements in accuracy and computational efficiency, underscoring the potential of this approach as a foundation for efficient and high-performance lip-reading systems. The proposed method highlights landmark-based modeling's potential as an efficient alternative to conventional image-based architectures. The following section describes the model's detailed architecture and implementation.

II. LITERATURE REVIEW

A. Image Based Lip-Reading Approaches

Most current lip-reading systems depend on image-based representations. In the past, CNN was used to obtain spatial features from lip regions, and then recurrent architectures, such as LSTM or Bi-GRU, were used to model temporal dependencies. For example, CNN and Bi-GRU architectures were used to sort 64 Bahasa words from lip region video frames [10]. Other studies used full-face images instead of just the lips, which made the data more

complex and required more processing power. Architectures such as the LRCN did a great job of recognizing things, but they had to use convolution processing to do it. Most VSR systems that use deep learning follow a pipeline that includes input preparation, spatial feature extraction, time modeling, and data classification [22]. 2D or 3D convolutional neural networks (CNNs) are usually used to extract spatial features, and recurrent networks or transformer-based architectures are used to model temporal dependencies [22], [23]. Even though these methods work well, they often cost a lot to run because they have to deal with high-dimensional pixel data.

B. Landmark Based Lip Analysis

Landmark-based methods attempt to reduce the number of dimensions in the input by using key points to show how the lip moves. In the past, researchers looked at how well people could recognize vowels by following four important lip landmarks [24]. A later study used the Dlib framework with 64 facial landmarks to make it easier to recognize SIBI [25]. In numerous instances, landmarks serve primarily as supplementary features or preprocessing steps, whereas the principal modeling pipeline is image-based [2], [26]. Recent surveys have shown that there is a growing need for VSR models that can be run on computers that do not have a lot of resources [22]. LSTM and BiLSTM are still very popular for temporal modeling, and attention mechanisms have been shown to improve sequence learning by highlighting important features across frames [23], [27]. Despite these advancements, the utilization of lip landmarks as a principal compact input representation for sequential modeling remains quite restricted.

C. Bahasa Lip-Reading and Dataset Limitation

Research specifically targeting lip-reading in Bahasa remains limited compared with that in English and other widely studied languages. Existing studies have predominantly focused on vowel recognition or have utilized relatively small datasets [1], [8], [10]. The scarcity of large-scale and standardized audio-visual datasets is a major challenge. The LUMINA dataset contains 15,301 videos from 14 speakers covering 2,931 Bahasa words [1]. However, LUMINA provides facial data cropped to the region between the nose and the chin, which restricts its applicability for reliable landmark extraction using MediaPipe Facemesh [28], [29]. This framework requires full-face input to establish stable geometric references for landmark normalization, as landmarks are estimated based on global facial geometry rather than isolated mouth regions. Consequently, partial facial input may lead to inconsistent lip landmark trajectories, making the proposed landmark-based pipeline unsuitable. In contrast, the IndoLR dataset comprises 5,807 videos from eight speakers and includes grouped word and phrase classifications [14], [30]. It has demonstrated strong performance in image-based lip-reading tasks. However, its scalability to larger and more diverse datasets, such as LUMINA, remains an open question. This highlights the importance of developing efficient approaches tailored to underrepresented languages, such as Bahasa.

Although image-based models and landmark extraction methods have been explored in previous studies, the use of lip landmarks as a primary compact input representation for efficient Bahasa lip-reading remains limited. Reducing computational complexity while maintaining competitive accuracy has received relatively little attention. These gaps motivate the proposed lightweight Attention-BiLSTM framework.

III. METHODS

The proposed approach follows a structured experimental pipeline comprising video frame extraction, lip landmark detection and normalization, sequential modeling using an Attention-BiLSTM architecture, and final class prediction. The overall workflow of this process is illustrated in Fig. 1. The following subsections describe each stage of the methodology in detail.

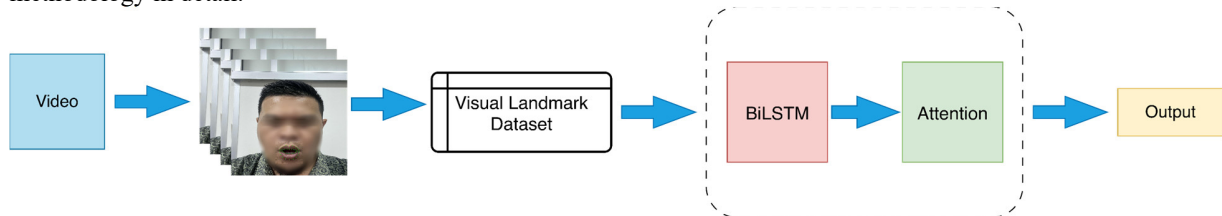


Fig. 1 Research stage

A. Dataset Description

The datasets referenced in this study are summarized in Table 1 to enhance clarity and facilitate structured comparison. IndoLR serves as the primary dataset for experimental evaluation, while MIRACL-VC1 is cited to support the proposed architectural framework's prior validation. The IndoLR dataset consists of 5,807 video samples collected from eight speakers, comprising expressions commonly used in everyday Bahasa communication. It includes

10 word classes: “*maaf*” (sorry), “*tolong*” (please), “*permisi*” (excuse me), “*halo*” (hello), “*mulai*” (start), “*berhenti*” (stop), “*lanjut*” (next), “*sakit*” (hurt), “*kembali*” (back), and “*awas*” (be careful). Additionally, it features 4 phrase classes: “*terimakasih*” (thank you), “*minta tolong*” (please help), “*saya minta maaf*” (I am sorry), and “*saya minta tolong*” (I am asking for help). MIRACL-VC1 is an English visual speech data set referenced in our previous work. Unlike IndoLR, which is provided in video format, this data set consists of 10 word classes and contains preextracted frame sequences.

TABLE 1
DATASET REFERENCED IN THIS STUDY

Dataset	Language	Classes	Modality	Role in Study
IndoLR	Bahasa	10 words, 4 phrases	Audio-Visual	Primary dataset for training and evaluation
MIRACL-VC1	English	10 words	Visual	Referenced for prior experimental validation

The original IndoLR dataset is organized into eight primary folders that correspond to the speakers. Each folder contains 14 subfolder representing 10 word classes and 4 phrase classes. To ensure that classification focuses on linguistic content rather than speaker identity, the dataset was reorganized by grouping samples according to speech class. The final dataset was divided into training, validation, and testing sets with a ratio of 80:10:10. The number of samples per category ranges from approximately 372 to 420 videos across all word and phrase classes, indicating a relatively balanced class distribution. This balanced structure helps mitigate potential class imbalance during training.

B. Data Preprocessing

First, the video samples were converted into individual frames to generate sequential image data [31], [32]. A fixed-length frame selection strategy was applied to ensure consistency across samples. For word-level classification, 40 frames were selected per sample, whereas 60 frames were used for phrase-level classification. These sequence lengths were chosen to provide a richer temporal representation compared with those of previous studies that used shorter sequences [33], [34]. When a video contained more frames than required, the central frames were selected to preserve the temporal symmetry. If the number of available frames was insufficient, the last frame was duplicated until the required sequence length was reached.

Lip landmark detection was performed following frame extraction using the MediaPipe Facemesh framework [35], [36], [37]. From the detected facial landmarks, 40 lip-specific points were selected to represent lip motion. The 2D (x, y) coordinates of these landmarks were extracted for each frame, resulting in a compact representation of lip movements. Therefore, each sample was represented as a sequence of normalized landmark coordinates with dimensions (40, 80) for word classes and (60, 80) for phrase classes, where 80 corresponds to 40 landmarks for 2 coordinate values. Both the proposed model and the LRCN-3Conv baseline employ MediaPipe to find the lip region in the same raw video frames to ensure fair comparison. The image-based model processes the cropped lip patches as pixel inputs, whereas the proposed method converts the same region into normalized landmark coordinate sequences. Thus, performance differences are attributable to feature representation and architectural design rather than preprocessing variations.

C. Lip-Landmark Extraction and Normalization

Before performing lip landmark extraction, the facial landmark detection framework employed in this study must be briefly described. The system detects 468 facial landmarks distributed across the entire facial region [38], [39], [40], [41]. These landmarks are defined within a structured mesh representation that models global facial geometry, enabling stable tracking of facial movements across frames. Alternative region-based detection methods, such as the Viola–Jones method, primarily localize facial areas but do not provide the dense and temporally consistent landmark representations required for sequential modeling [42], [43].



Fig. 2 Detected lip landmarks

As illustrated in Fig. 2, the selected lip landmarks are overlaid on each frame to identify key positions in the lip region. Lip landmarks can be more accurately positioned when the face is fully visible, as the lip area can be clearly distinguished from other facial features, such as the eyes, nose, and forehead. However, the placement of lip landmarks becomes inconsistent when parts of the face are missing or occluded, as observed in the LUMINA dataset. Based on the experiments conducted in this study, some landmarks remain correctly positioned on the lip region, whereas others are incorrectly placed on non-lip facial areas. Therefore, complete and properly captured facial images are required to achieve optimal results. The coordinates of these landmarks were extracted and organized sequentially for each frame according to their original video source, ensuring temporal alignment between frames. The scale of the coordinates of each frame may vary due to inconsistencies in the speaker's lip position.

Normalization is crucial because the raw landmark coordinates x and y depend on the image size or the dimensions of the bounding box, both of which can fluctuate significantly [44]. Without normalization, it is challenging for the model to recognize patterns. For instance, in images of different sizes (300×300 pixels and 600×600 pixels), a lip landmark positioned at the center would have coordinates (150, 150) and (300, 300) in the first and second images, respectively. If left unnormalized, the model would incorrectly interpret these as distinct locations despite their identical relative positions. The y coordinates are normalized to a fixed range of 0 to 1 to resolve this issue. Rather than relying on absolute pixel values, this process involves identifying the minimum and maximum coordinate values to define the lip region within the bounding box.

Normalization is achieved by linearly translating the original coordinates into the lip region's border box range. The smallest and largest values of the bounding box are expressed as *left_bound*, *right_bound*, *top_bound*, and *bottom_bound*, respectively, and are used to determine the area boundary in the size adjustment process. The normalized horizontal and vertical coordinates, X_{norm} and Y_{norm} , are computed using Equations (1) and (2), respectively, resulting in normalized landmark (x, y) representations suitable for subsequent model training. Without this normalization step, variations in face distance, image resolution, and lip region size introduce scale inconsistencies across samples, potentially degrading model generalization and increasing classification error.

$$X_{norm} = \frac{x - \text{left_bound}}{\text{right_bound} - \text{left_bound}} \quad (1)$$

$$Y_{norm} = \frac{y - \text{top_bound}}{\text{bottom_bound} - \text{top_bound}} \quad (2)$$

D. Model Architecture

The proposed architecture is designed to model the temporal dependencies of lip landmark sequences. While convolution-based architectures, such as LRCN-3Conv, extract high-dimensional spatial features from image pixels using stacked CNN layers followed by a unidirectional LSTM, the proposed approach directly models sequential landmark representations [14]. As shown in Fig. 3, the model consists of a BiLSTM layer followed by an attention mechanism inspired by the additive attention formulation of Bahdanau et al. [45], [46]. LSTM networks mitigate vanishing gradient issues through gated memory structures, whereas BiLSTM processes sequences in both forward and backward directions to provide richer contextual representations [47], [48], [49], [50]. The input tensor for word-level classification is organized as (batch_size, 40, 80), representing the sequence length and flattened landmark features. The BiLSTM layer uses a hidden size of 64 per direction, resulting in hidden representations H_t with dimension 128 at each timestep. For phrase-level classification, the hidden size was increased to 256 to capture more complex temporal dependencies [51].

$$\alpha_t = \frac{\exp(W_{i,j} H_t + b_j)}{\sum_{k=1}^T \exp(W_{i,k} H_k + b_k)} \quad (3)$$

$$\text{context} = \sum_t \alpha_t H_t \quad (4)$$

The attention mechanism computes attention weights α_t according to Equation (3). In this formulation, $W_{i,j}$ denotes the learnable weight matrix that linearly transforms each hidden state H_t , while b_j represents the corresponding bias term. These parameters are shared across all timesteps, implying that the same linear transformation is applied to each hidden state of BiLSTM. The resulting attention scores are normalized using the softmax function across the temporal dimension, producing attention weights α_t that sum to one. Then, the context vector is calculated using Equation (4), where each hidden state H_t is weighted by its corresponding attention weight α_t . This operation enables the model to emphasize informative frames within each lip movement sequence while reducing the influence of less relevant temporal segments. Subsequently, the final context representation is passed to a linear classification layer to produce

logits corresponding to the target classes. The model is trained using the cross-entropy loss function for multiclass classification.

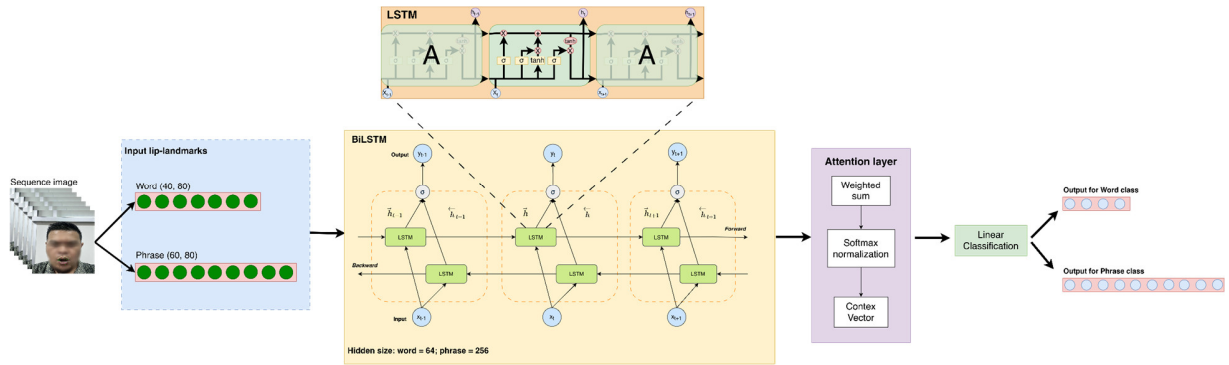


Fig. 3 Model architecture

E. Training Configuration

Model optimization was performed using the designated training, validation, and testing splits. Optimization was performed using the Adam optimizer with a learning rate of 0.005 and a weight decay of 1×10^{-5} to balance convergence and regularization. A dropout rate of 0.5 was applied during training to mitigate overfitting. Both word-level and phrase-level models were trained for 100 epochs using a batch size of 32. The best-performing model for each task was selected based on validation accuracy. The best-performing model was selected on the basis of validation accuracy. All experiments were implemented in PyTorch and executed on the GPU-accelerated Google Colab Pro environment with GPU acceleration.

F. Ablation Study Design

An ablation study was conducted on the IndoLR data set to systematically evaluate the contribution of architectural components within the proposed landmark-based framework. Under identical training conditions, four recurrent configurations were examined: (1) LSTM, (2) BiLSTM, (3) BiGRU, and (4) BiGRU-Attention [3], [52], [53]. All models were trained using the same train-validation-test split, identical hyperparameter settings, and the same landmark preprocessing pipeline to ensure methodological consistency. By maintaining uniform training configurations across all variants, any observed differences in performance can be directly attributed to architectural design choices rather than discrepancies in optimization strategy or data preparation. This controlled setup enables a reliable evaluation of how recurrent directionality (RD) and attention mechanisms influence recognition performance within the landmark-based paradigm.

G. Evaluation Matrix

The proposed model evaluates the performance of the proposed model, and an assessment will be conducted focusing on both classification accuracy and computational efficiency. Classification performance will be quantified using standard metrics, including accuracy, precision, recall, and F1 score, which collectively provide a robust understanding of the model's capability to accurately and comprehensively identify each class. This comprehensive evaluation will ensure that the model is effective in terms of accuracy, processing speed, and computational cost. Four key metrics are applied to evaluate the performance of the word and phrase datasets: accuracy (Equation 5), precision (Equation 6), recall (Equation 7), and F1 score (Equation 8). These measurements are based on four outcomes: a TP occurs when a test correctly detects the existence of a condition, whereas a TN represents a test result that accurately predicts its absence. Conversely, a false positive (FP) is an error where the presence of an attribute is falsely suggested, and a false negative (FN) occurs when a test incorrectly implies that a particular circumstance is not present.

Simultaneously, computational factors such as the average inference time (It) per sample, the number of trainable parameters, and GFLOPs (Giga Floating-Point Operations) are analyzed to assess the suitability of the model for real-time applications and resource-constrained environments [54], [55]. The inference time was measured at the model level, excluding shared preprocessing stages to ensure a fair comparison. This ensures that the reported inference time reflects architectural design differences rather than preprocessing pipeline variations. GFLOPs quantify the computational cost of a model by representing the total number of floating-point operations required for inference,

expressed in billions [56]. A lower value signifies a streamlined, lightweight architecture, whereas higher values indicate greater computational demands. As shown in Equation (9), the GFLOPs are calculated by normalizing the sum of floating-point operations from the BiLSTM (F_{BiLSTM}), Attention ($F_{Attention}$), and fully connected (F_{FC}) components by a factor of 10^9 .

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (5)$$

$$Precision = \frac{TP}{TP+FP} \quad (6)$$

$$Recall = \frac{TP}{TP+FN} \quad (7)$$

$$F1 \text{ score} = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (8)$$

$$GFLOPs = \frac{F_{BiLSTM} + F_{Attention} + F_{FC}}{10^9} \quad (9)$$

IV. RESULTS

The image-based model shows a rapid increase in training accuracy, reaching values close to 100% within a relatively small number of epochs, in terms of training dynamics. This pattern indicates that the model can quickly fit the training data. However, the corresponding training loss curve still exhibits noticeable fluctuations during the optimization process, suggesting that despite the high training accuracy, the learning process is less stable. In contrast, the landmark-based model demonstrates a more gradual increase in training accuracy, accompanied by a more consistent decrease in training loss. This trend suggests a more stable optimization process and smoother convergence across epochs. These training behaviors are illustrated in Figs. 4(a) and 4(b), which present the training accuracy and training loss curves, respectively.

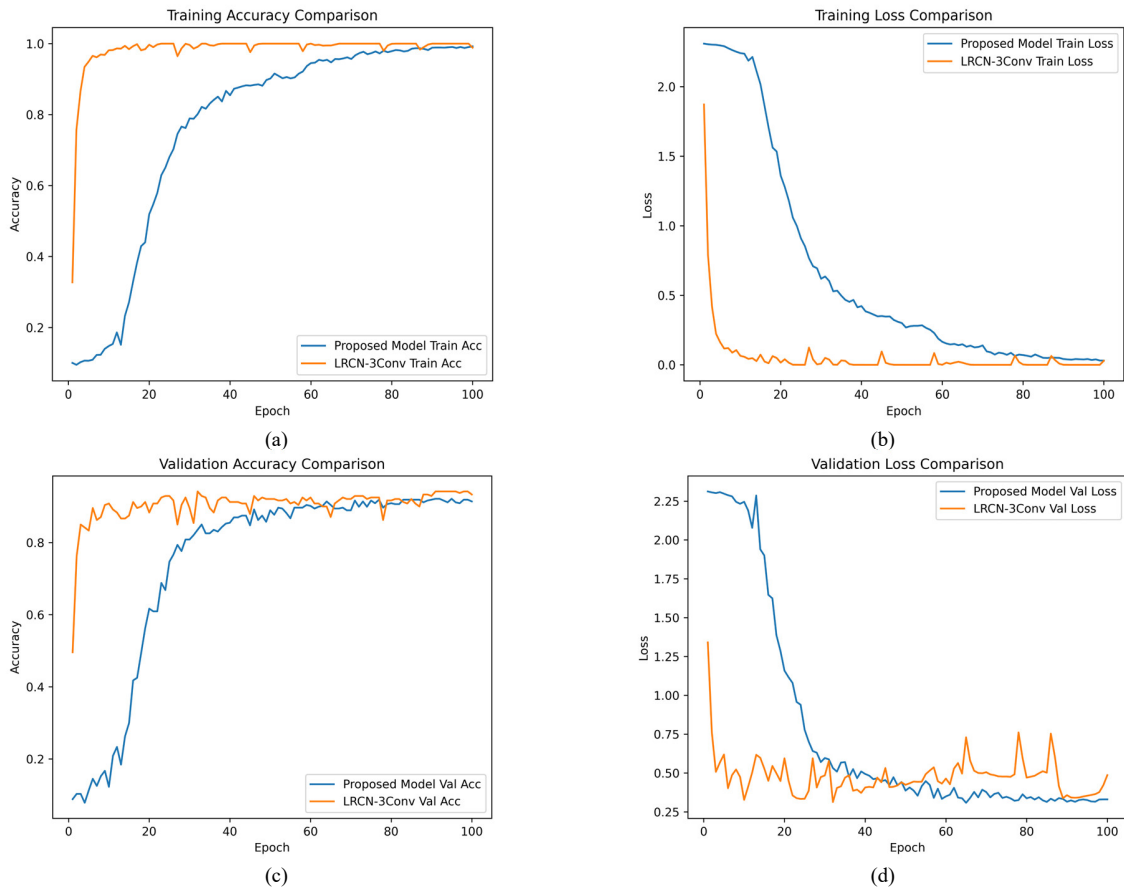


Fig. 4 Training and validation performance of the proposed landmark-based model for word-level lip-reading classification: (a) Training accuracy; (b) Training loss; (c) Validation accuracy; (d) Validation loss across 100 training epochs

A. Training and Validation Accuracy

The proposed landmark-based model achieves a maximum validation accuracy of 92.28% in the word classification scenario. This performance occurs within a competitive range when compared to the image-based LRCN-3Conv model reported in previous studies, which attained the best validation accuracy of 97.92% on the same dataset. Although a performance gap exists, the proposed model can maintain a high level of accuracy while relying on a significantly more compact input representation. The proposed model is substantially simpler than the LRCN-3Conv architecture in terms of model complexity. It employs significantly fewer learnable parameters at both the word- and phrase-level tasks, primarily because it avoids convolution-heavy image processing (CHP). This reduction is mainly attributed to the use of lip landmark coordinates instead of convolution-heavy image processing. Therefore, the proposed approach offers a favorable trade-off between recognition performance and computational efficiency. The validation curves for word classification, shown in Figs. 4(c) and 4(d), further indicate that the landmark-based model experiences a steady increase in validation accuracy until convergence. The consistent decrease in validation loss and the controlled gap between training and validation accuracy also support this trend, suggesting stable learning behavior with no strong indication of overfitting during training.

The proposed landmark-based model obtained an overall accuracy of 95% on the phrase classification test, and the macro precision, recall, and F1-score values all remained at 0.95%. These results demonstrate that the proposed technique effectively captures extended temporal relationships essential for modeling increasingly complex language units. By comparison, previous studies have shown that the image-based LRCN-3Conv model has an average accuracy, recall, and F1-score values of approximately 95%. Despite relying on a fundamentally different and more compact input representation, the proposed landmark-based approach achieves comparable performance to the image-based baseline, indicating that similar recognition accuracy can be maintained with reduced input complexity.

The stability of this landmark-based model in the training and validation process is attributable to a more concentrated representation of inputs with reduced dimensions. Using lip landmarks as input data can help reduce the noise and variation that is common in the image, which makes the optimization process more controlled. Additionally, the pre-processing step performed on landmark data helps to lower the risk of overfitting by using structured geometric representations. This allows the model to learn general lip movement patterns. The BiLSTM architecture and the attention mechanism work together to make the model focus on the most important time segments, which helps it converge more consistently. This method leads to a more stable training and validation curve than image-based models because the model's complexity is balanced by the data size.

B. Computational Complexity

The proposed method demonstrates a substantially lower computational complexity than previous image-based approaches. It requires only 0.01 GFLOPs for word classification and 0.14 GFLOPs for phrase classification, whereas the LRCN-3Conv model requires 0.62 and 0.83 GFLOPs. After 100 training epochs, the model achieves 93.03% test accuracy for word classification and 95.18% for phrase classification. Table 2 provides a detailed comparison with previous studies. These results confirm that lip landmark representations can effectively capture the distinctive motion patterns required for accurate recognition while reducing computational demands. Although the proposed model achieves slightly lower accuracy than image-based architectures, it does so with markedly reduced computational complexity. This reduction supports its suitability for deployment on resource-constrained platforms such as mobile or embedded systems, enabling practical real-world lip-reading applications.

TABLE 2
MODEL PERFORMANCE COMPARISON FOR WORDS AND PHRASE CLASS

Model	Class	Val.acc (%)	Test.acc (%)	It (s)	Parameter	GFLOPs (s)
Proposed model	Words	92.28	93.03	0.000530	175499	0.01
	Phrase	97.57	95.18	0.011366	2271749	0.14
LRCN-3Conv	Words	97.92	95.42	0.211583	3299178	0.62
	Phrase	99.37	95.63	0.271884	3298788	0.83

This study used precision, recall, and F1-score metrics to further test the effectiveness of the attention-based BiLSTM model for classifying both words and phrases. The classification report for the word-level task has an overall accuracy of 93% across 10 word classes (Table 3). The macro and weighted averages for all evaluation metrics are also 93%. The evaluation results in Table 4 show that the model correctly classifies the four phrase classes, with an overall accuracy of 95% on the test data set. These results show that the proposed model is very strong at both word- and phrase-level lip-reading tasks. They also showed that it could be useful in real-world translation and assistive communication systems.

TABLE 3
COMPARISON MODEL PERFORMANCE EVALUATION FOR WORDS CLASS

Words	Precision (%)	Proposed model	F1 Score (%)	Precision (%)	LRCN-3Conv	F1 Score (%)
		Recall (%)			Recall (%)	
<i>Awas</i>	0.92	1.00	0.96	0.96	1.00	0.98
<i>Berhenti</i>	0.90	0.90	0.90	0.96	1.00	0.98
<i>Halo</i>	0.88	1.00	0.94	0.91	0.88	0.89
<i>Kembali</i>	0.93	1.00	0.97	0.96	0.96	0.96
<i>Lanjut</i>	0.87	0.85	0.86	0.92	0.92	0.90
<i>Maaf</i>	1.00	0.88	0.93	1.00	1.00	1.00
<i>Mulai</i>	0.96	0.98	0.97	0.92	0.92	0.94
<i>Permis</i>	0.98	0.96	0.97	0.96	0.96	0.98
<i>Sakit</i>	0.93	0.88	0.90	0.96	0.96	0.96
<i>Tolong</i>	0.90	0.85	0.88	0.96	0.96	0.96
Accuracy			0.93			0.95
Macro avg	0.93	0.93	0.93	0.95	0.95	0.95
Weighted avg	0.93	0.93	0.93	0.95	0.95	0.95

TABLE 4
COMPARISON MODEL PERFORMANCE EVALUATION FOR PHRASE CLASS

Phrase	Precision (%)	Proposed model	F1 Score (%)	Precision (%)	LRCN-3Conv	F1 Score (%)
		Recall (%)			Recall (%)	
<i>Minta Tolong</i>	0.95	1.00	0.97	0.93	0.95	0.94
<i>Saya Minta Maaf</i>	1.00	0.90	0.95	0.97	0.97	0.97
<i>Saya Minta Tolong</i>	0.92	1.00	0.96	0.93	0.93	0.93
<i>Terimakasih</i>	0.96	0.91	0.93	1.00	0.97	0.99
Accuracy			0.95			0.95
Macro avg	0.95	0.93	0.95	0.95	0.95	0.95
Weighted avg	0.95	0.93	0.95	0.95	0.95	0.95

C. Ablation Study

An ablation study was conducted on the IndoLR dataset to investigate the contribution of recurrent directionality and attention mechanisms within the landmark-based framework for both word and phrase level tasks. The results are summarized in Table 5. At the word level, incorporating attention improves recognition performance compared with recurrent models without attention. The BiGRU-Attention configuration achieved the highest test accuracy (95.10%), followed by BiGRU (92.65%), LSTM (91.67%), and BiLSTM (88.97%). The performance gap between models without attention indicates that simple last-timestep pooling is less effective in capturing discriminative temporal cues.

TABLE 5
ABLATION STUDY ON LANDMARK-BASED INDOLR WORD AND PHRASE CLASSIFICATION

Class	Model	Val.acc (%)	Test.acc (%)	It (s)	Parameter	GFLOPs
Words	BiGRU+Attention	92.14	95.10	0.000020	460555	0.018
	BiGRU	91.40	92.65	0.000013	460298	0.018
	LSTM	88.70	91.67	0.000011	240906	0.009
	BiLSTM	87.71	88.97	0.000013	612874	0.024
Phrase	BiGRU+Attention	96.36	94.58	0.000121	1704453	0.102
	BiGRU	32.12	34.94	0.000115	1703940	0.102
	BiLSTM	26.67	22.89	0.000119	2271236	0.136
	LSTM	26.67	22.89	0.000066	873476	0.052

The effect becomes more pronounced in phrase-level classification, where the model must include longer temporal dependencies. In the ablation setting, the BiGRU-Attention model achieved 94.58% test accuracy, whereas recurrent variants without attention exhibited substantial performance degradation (34.94% for BiGRU and 22.89% for both BiLSTM and LSTM). This substantial gap highlights the importance of STM when modeling extended linguistic units. Interestingly, while the BiGRU-Attention configuration achieved the highest accuracy at the word level, the proposed BiLSTM-Attention architecture demonstrates slightly stronger phrase-level performance in the primary experimental comparison (95.18% test accuracy, as shown in Table 2). This tendency suggests that the LSTM gating mechanism provides marginally improved capability in modeling longer temporal dependencies. However, the performance difference remains modest, indicating that both attention-based recurrent architectures are effective for extended sequence modeling (ESM).

V. DISCUSSION

This study presents an innovative lip-reading methodology for Bahasa, a relatively unexamined research area in comparison to more extensively researched languages. In contrast to previous works that primarily depend on image-based representations [5], [57], [58], this study utilizes lip landmark-based data [30] as the primary input representation. By concentrating on landmark features, the proposed method markedly diminishes input dimensionality while retaining the critical information necessary to capture lip movement dynamics. Consequently, the proposed model achieves enhanced computational efficiency without compromising recognition performance. The experimental results show that a BiLSTM with an attention mechanism, which is a relatively lightweight architecture, can still achieve high accuracy when trained on a smaller dataset. While research in VSR has grown globally, its application to the Bahasa remains markedly underdeveloped, particularly regarding the diversity of data modalities. Current studies utilizing the LUMINA dataset predominantly rely on raw pixel-based features for either text transcription or speech-to-lip synthesis, leaving the potential of facial landmarks, which offer a more compact, noise-resistant, and geometrically structured representation of speech, largely unexplored in the Indonesian context [59], [60].

A quantitative comparison with the image-based LRCN-3Conv model provides further insight into this trade-off [14]. Although the image-based model achieves slightly higher accuracy at both word and phrase levels, the performance gap remains relatively small when viewed in aggregate metrics. However, this marginal gain comes with a substantially higher computational cost, as reflected in the image-based architecture's significantly greater GFLOP requirements. The results indicate that the proposed landmark-based model maintains competitive precision, recall, and F1-scores across word and phrase categories, demonstrating stable recognition performance. The integration of BiLSTM with an attention mechanism effectively captures temporal dependencies, allowing the model to compensate for the absence of high-dimensional RGB inputs. This suggests that performance improvements in image-based models are not solely attributable to superior temporal modeling but are also influenced by higher input dimensionality. While the LRCN-3Conv architecture benefits from rich spatiotemporal representations derived from full-frame images, this representational richness increases the complexity of the model. In contrast, the landmark-based approach leverages structured geometric features to reduce redundancy while preserving essential motion information. The substantially lower GFLOP values confirm that the proposed method offers improved computational efficiency, making it more suitable for deployment in resource-constrained environments.

Although these results are promising, several limitations should be acknowledged. The developed model has not yet been evaluated on a larger dataset that includes a wider array of real-world variations, such as variations in speaker accents, lip articulation styles, and lighting conditions [17]. These factors might affect the strength and generalizability of the proposed approach, and future research should look into them. Although the comparison with LRCN-3Conv was conducted on the same dataset, differences in implementation details, hyperparameter tuning, or training procedures may introduce potential internal validity concerns. Furthermore, the effectiveness of the proposed method depends on the accuracy of the landmark extraction process [14]. Errors in landmark detection under challenging visual conditions could propagate to the classification stage, posing a threat to construct validity. Therefore, future research should validate the proposed framework on larger and more diverse datasets and perform fully controlled experimental comparisons.

Table 6 summarizes prior studies on visual speech recognition across datasets and data modalities. The results indicate that both image- and landmark-based approaches can achieve high accuracy, although image-based models tend to yield slightly higher performance. However, landmark-based methods demonstrate competitive results while operating with lower input dimensionality, highlighting a more efficient alternative for practical deployment.

TABLE 6
COMPARISON OF PRIOR STUDIES ON VISUAL SPEECH RECOGNITION AND SYTHESIS

Model	Dataset	Data Modality	Accuracy (%)
CSA-ResNET-34+BGM [62]	IDLRW	Lip images	60.51
BiGRU+Attention [63]	MIRACL-VC1	Lip landmark	81.75
BiLSTM with CTC [64]	MIRACL-VC1	Facial landmark	93.30
CNN+BiLSTM with Adapters [65]	LRW-ID	Lip landmark	92.33
Conv3D+BiLSTM [66]	GRID	Lip images	98.40

The advantages of the landmark-based approach are more evident in phrase-level classification, where longer and more complex temporal structures must be modeled. By integrating a BiLSTM with an attention mechanism, the model can capture long-range temporal dependencies by emphasizing the most informative segments within each sequence. This contributes to the consistently high performance across the phrase classes. Although the image-based

baseline achieves slightly higher F1-scores, the landmark-based method delivers comparable performance with substantially lower computational cost. These findings highlight a favorable trade-off between accuracy and efficiency, which is particularly relevant for real-time applications and resource-constrained environments.

To further mitigate potential dataset-specific bias, evidence from prior studies (Table 6) indicates that attention-based recurrent architectures consistently demonstrate strong performance across different datasets. This cross-dataset consistency suggests that temporal modeling with attention is not inherently language-dependent. Therefore, the performance observed in the IndoLR dataset is likely attributable to the modeling approach's robustness rather than dataset-specific characteristics.

VI. CONCLUSIONS

This study presents a lip-reading method specifically developed for Bahasa that uses lip landmarks to classify 10 word classes and 4 phrase classes. Concentrating on lip coordinates as the primary data source, this approach effectively generates a smaller dataset while accurately capturing sequential lip movements. This allows for the precise classification of speech into categories. Employing a BiLSTM model enhanced with an attention layer, the method achieves 93.03% and 95.18% accuracy for word and phrase classes, respectively. In addition to its high accuracy, this approach demonstrates efficiency in terms of both training time and computational resources. In addition, the average inference time per sample was measured at 0.000530 s for word classes and 0.011366 s for phrase classes. Furthermore, the model shows impressive efficiency, with GFLOPs of 0.01 and 0.14 for word and phrase classes, respectively, making it highly suitable for resource-constrained environments. These characteristics highlight the potential of lip landmarks as an effective alternative to more complex image-based datasets. The main contribution of this study is the development of a lightweight and efficient lip-reading model, paving the way for broader applications of lip-reading technology in Bahasa, including deployment on devices such as mobile phones with limited resources. However, further research is necessary to construct models for more diverse Bahasa datasets. Future studies can also explore practical applications in real-world scenarios, thereby enhancing the proposed method's practical value.

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Data Availability: The data supporting this research is openly available in kaggle.com at <https://www.kaggle.com/datasets/abasset/indolr> [30].

Informed Consent: The data used is from the IndoLR Dataset, an open-source corpus for Lip-Reading research. The original data sources handled the initial ethical consent. The authors did not acquire data from new human participants and solely processed extracted features from this available dataset.

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